

Text-to-Speech (TTS)

Synthesis: Basics and Models

Introduction to TTS

- Definition:
- Text-to-Speech (TTS) synthesis is the process of converting written text into spoken words using computer programs. It enables machines to "speak" like humans.
- Applications of TTS:
 - Assistive technologies for visually impaired users (screen readers)
 - Virtual assistants (Alexa, Google Assistant, Siri)
 - Navigation systems
 - Language learning tools
 - Audiobooks

Key Goals of TTS Systems:

- Naturalness: Voice should sound human-like.
- Intelligibility: Speech must be clear and understandable.
- Expressiveness: Optional features like emotions, pitch, and style.

Basic Architecture of a TTS System

- A typical TTS system consists of two main stages:
- Stage 1: Text Analysis / Front-End
- Stage 2: Speech Synthesis / Back-End

Stage 1: Text Analysis / Front-End

- Converts raw text into a linguistic representation.
- Steps include:
 - Text normalization: Expanding numbers, dates, abbreviations.
 - e.g., "Dr. Smith will arrive at 9 a.m." → "Doctor Smith will arrive at nine in the morning."
 - Tokenization: Breaking text into words and sentences.
 - Part-of-speech (POS) tagging: Identifying nouns, verbs, etc.
 - Syntactic and semantic analysis: Helps with correct pronunciation and prosody.
- Output: A phonetic transcription with prosody features (intonation, stress, duration).

Stage 2: Speech Synthesis / Back-End

- Converts phonetic and prosodic representation into audio signals.
- Methods include:
 - **Concatenative Synthesis**
 - Uses a large database of recorded speech units.
 - Units: phonemes, diphones, syllables, or words.
 - Process: Select the best matching units → concatenate → generate speech.
 - Pros: High naturalness.
 - Cons: Limited flexibility, large storage requirement.

Stage 2: Speech Synthesis / Back-End cont...

- **Formant Synthesis**

- Uses acoustic models of human vocal tract instead of recordings.
 - Generates speech by synthesizing formants (resonant frequencies of speech sounds).
- Pros: Small size, flexible.
- Cons: Speech sounds robotic and less natural.

Stage 2: Speech Synthesis / Back-End cont...

- **Articulatory Synthesis**

- Simulates physical vocal tract movements to produce speech.
 - Very complex and research-intensive.
- Pros: Highly accurate modeling.
- Cons: Computationally expensive, less practical.

Stage 2: Speech Synthesis / Back-End cont...

- **Neural Network-Based Synthesis (Deep Learning TTS)**
 - Modern approach using neural networks to model speech.
 - Popular models:
 - Tacotron / Tacotron 2: Sequence-to-sequence model with attention.
 - WaveNet: Generates raw audio waveform sample-by-sample.
 - FastSpeech / FastSpeech 2: Improves speed and stability over Tacotron.
 - VITS (Variational Inference TTS): End-to-end TTS with natural prosody.
- Pros: Highly natural, expressive, and can adapt to different voices.
- Cons: Requires large datasets and GPU computation.

Features of High-Quality TTS

- Pitch (F0): Frequency of speech; used to create intonation.
 - Duration: Time length of phonemes.
 - Intensity / Loudness: Energy of speech for emphasis.
 - Voice Quality: Gender, accent, style, emotional tone.
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- **Prosody generation (intonation, rhythm, stress) is crucial for natural-sounding speech.**

Evaluation of TTS Systems

- Objective Metrics
 - Signal-to-noise ratio (SNR)
 - Mel-Cepstral Distortion (MCD)
 - Word error rate in recognition-based evaluation
- Subjective Metrics
 - MOS (Mean Opinion Score): Human listeners rate naturalness on a scale (1–5).
 - Intelligibility and naturalness tests.

Modern Trends in TTS

- Voice Cloning: Generate speech in a target person's voice.
- Multilingual TTS: Supports multiple languages and code-switching.
- Expressive / Emotional TTS: Captures emotions like happiness, sadness, anger.
- End-to-End Models: Reduce the need for feature engineering; direct text-to-waveform generation.

Example: Pipeline of a Neural TTS System (Tacotron 2)

- Input Text → text embedding
- Encoder: Converts text into latent representation
- Attention Mechanism: Aligns text with speech
- Decoder: Generates mel-spectrogram
- Vocoder (e.g., WaveNet): Converts mel-spectrogram to waveform
- Output: Natural-sounding audio speech

Summary

Aspect	Traditional TTS	Neural TTS
Data Requirement	Large pre-recorded units	Large text-audio pairs
Flexibility	Low (limited to recorded units)	High (can generalize to new text)
Naturalness	Medium to High	Very High
Prosody	Limited	Captures natural prosody
Computational Cost	Low	High